

(Ir)rationality

Javier Larrosa

UPC Barcelona Tech

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Introduction

We usually identify two cognitive modes in our brain:

Rationality

Believe something by: i) the identification of **relevant true premises** and ii) use them to reach **Sound Conclusions**

Intuition

Believe something from **instinctive feeling** rather than conscious reasoning

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Rationality seems to be superior.
Rationality seems a very human capability.
Humans like to think about ourselves as very rational.

The Rational Agent Assumption

Rational Agent Assumption

- People are generally rational, and their thinking is normally sound
- Emotions such as fear, affection, and hatred explain most of the occasions on which people depart from rationality (lost of control).
- This assumption has been made during centuries.
- It has influenced our understanding of the world. Most notably in Philosophy, all Social Sciences and Economics
- But it is being **revised** by Cognitive Scientists (Psychologists, Neuroscientists,...)

A More Realistic Model for the Mind

Watch the video in this link (Mandatory!!)

Thinking Fast and Slow (Kahneman 2002)

We have two thinking modes:

- **Fast Thinking:** Intuitive, effortless, automatic, inaccurate
 - Great for survival
 - Driven by fear, group acceptance,
 - Driven by efficiency: laziness, world simplification,...
 - Active most of the time
- **Slow Thinking:** Rational, tiring, self-awareness, more accurate.
 - Probably more recent in our evolution path
 - Takes the control sometimes
 - Sense of agency, selective attention.

Puzzle

- A **pencil** and a **eraser** cost \$1.10
- The **pencil** costs one dollar more than the **eraser**
- How much does the **eraser** cost?

we cannot avoid having our mind giving us a wrong quick answer

Puzzle

$$123 \times 407 =$$

What about that?

Puzzle

$$512 \div 16 =$$

In you are a Computer Scientist may be your fast thinking brain gives you a hint that is very useful to your slow thinking brain

Example

Fast Thinking: <https://www.youtube.com/watch?v=MEWSjzw2jVw>

Slow Thinking: <https://www.youtube.com/watch?v=vJG698U2Mvo>

An evolutionary explanation

Our Timeline

- 4000,000,000 Life
- 250,000,000 Mammals
- 6,000,000 Primates
- 200,000 Homo Sapiens
- 12,000 Agriculture, language, numbers, money
- 2,000 Philosophy (what can be known?)
- 400 Modern Science (enlightment)
- 200 French Revolution/American Revolution
- 75 Declaration of Human Rights

An evolutionary explanation

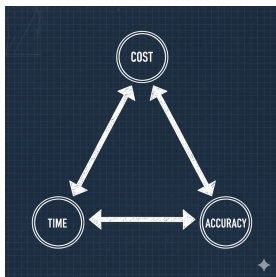
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Rational, deliberate (slow) thinking is very recent in our evolution line

The trade offs of our reasoning

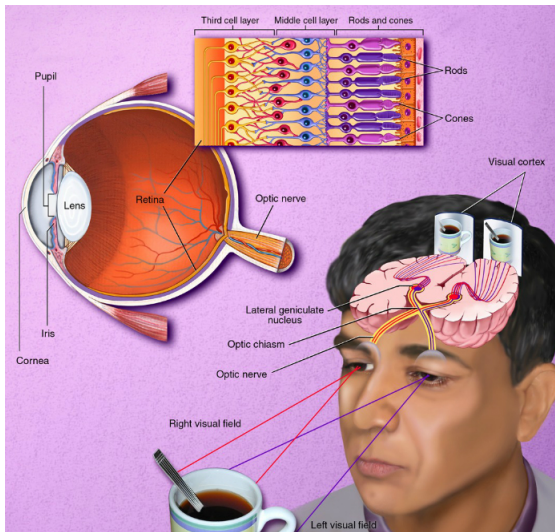
Evolution had to deal with a tension among three goals:



- The **Cost** of the cognitive process
- The **Speed** of the cognitive process
- The **Accuracy** of the cognitive process

The trade offs of our reasoning

For example, our vision (image recognition) is costly, fast and not very but sufficiently accurate.



Rationality and Intuition in Research

- Doing Research and/or Innovation requires **Rationality**
 - Observing the world with **objectivity**
 - Finding the **right issues** to address
 - Doing the **right analysis** to take the right course of action
- In fact, Research and Innovation requires a good balance of **Rationality** and **Intuition**
 - Newton's apple
 - Archimedes Eureka

What does it have to do with my (research) life?

Intuition vs Rationality

Research needs of both intuition and rationality. Due to evolution, our brain is a little unbalanced.

- Exercise your rationality: Critical Thinking (we will cover that in the next lectures)
- Be aware of your limitations (we will cover that in the rest of this lecture)

- Scientists have identified many forms of **flawed perception, reasoning and decision making**
- They are called **Cognitive bias**
- **Cognitive Bias** are well identified cases of **predictable irrationality**
- The reasoning errors associated to them are due to the design of our cognition machinery rather than the corruption of thought by emotion (our brain is buggy)
- They happen all the time in our life.
- It is very hard for us to fight against them.

The Rational Agent Assumption in Economics

Rational Agent Assumption in Economics (a.k.a homo economicus)

We always choose to perform the action with the optimal expected outcome for ourselves from among all feasible actions.

It is used to capture many fundamental Economic Principles:

- Price mechanisms

Are we really rational agents?

See details in [this link](#)

The Ultimatum Game

Two people X and Y (that do not know each other) have to split \$10 according to the following steps:

- 1 X decides how the money is split
- 2 Y accepts the deal or rejects it. In case of rejection they do not get anything

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The Ultimatum Game

Two people X and Y (that do not know each other) have to split \$10 according to the following steps:

- ① X decides how the money is split
 - ② Y accepts the deal or rejects it. In case of rejection they do not get anything
- What the rational agent assumption says? X does a 9-1 split and Y accepts it.
 - What happens in practice?
 - This is a Laboratory Experiment (limited significance)

Are we really rational agents?

Another experiment

An (typically) empty dining room in a corporation building with a coffee maker and a "honesty box"

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The Short-term vs Long-term self conflict:

- Being short-term rational means cheating
- Being long-term rational (in many cases) means not cheating

Are we really rational agents?

New scenario

A picture like that in front of the "honesty box"



Are we really rational agents?

New scenario

A picture like that in front of the "honesty box"



People cheat less with the photo

Irrationality due to our Social nature

The mere presence effect:

Not being alone changes our cognitive processes

Being with people makes us slightly smarter (if we do not feel pressure) or dumber (if we feel pressure)

Also observed in chimpanzees solving cognitive tasks

Truth Telling (Abeler et al., 2012)

- Phone call asking to toss a coin.
- If heads 15 Euros immediately transferred.
- Almost all participants report their outcome honestly

See details in [this link](#)

What does it have to do with my (research) life?

Hint: take advantage of our "social" irrationality

- Write a "social" contract
- Put (maybe virtual) eyes supervising you

Irrationality assessing likelihood of events

Assign probabilities to events

- 1 Flooding in the US in the next 10 years killing at least 1000 people
- 2 Earthquake in California in the next 10 years causing a flooding that kills at least 1000 people

Assign probabilities to events

How much would you be willing to pay for an insurance that pays you \$100,000 in case of:

- 1 Dead caused by unnatural causes
- 2 Dead caused by a terrorist incident

Insurance companies exploit that

See details in this link

Grading Pleasure

Take two wines made out of the same type of grape. One is cheap (\$14, simple production) and the other is expensive (\$130 complex production).

- 1 Grade them without knowing the price (small difference)
- 2 Grade them knowing the price (larger difference)
- 3 Monitor enjoyment (brain sensors) without knowing price (small difference)
- 4 Monitor enjoyment (brain sensors) knowing price (larger difference)

Luxury car brands make profit out of that

Clustering Illusion

During WWII, londoners developed complex theories about the pattern of bomb impacts within London. But a statistical analysis shows that they were almost random.

Casinos exploit this

Hot Hand Phenomenon

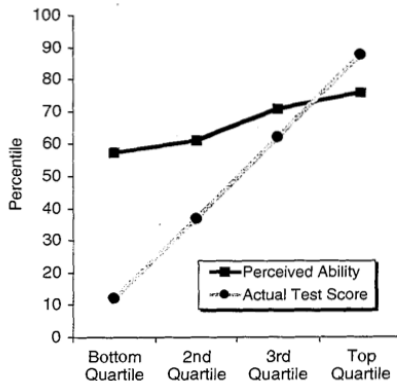
You are the coach of a Basketball team. During a game, one second before the end, there is a tie. Your team has two free shots. Who should you assign the shooting?

- 1 A player that has scored 10 free shots during the game without missing any.
- 2 The player with the best free shots average last month
- 3 The player with the best free shots average along the season.

Irrationality due to our self-confidence

The Dunning-Kruger Effect

People of low ability tend to overestimate their abilities.



This is an old experiment. [See link for details](#)

See details in this link

How politics makes us stupid, Kahan 2013

Researchers are experimenting with a new cream for skin rash. Consider the results of this experiment

	Improve	No Improve
Cream	223	75
No Cream	107	21

Please indicate whether the experiment shows that using the new cream is likely to make the skin condition better or worse.

Results support the *Science Comprehension Thesis*: the better subjects were at math, the more likely they were to stop, work through the evidence, and find the right answer.

How politics makes us stupid, Kahan 2013

Politicians are studying the effect of banning guns on crime rate. Consider the following data

	Improve Crime rates	No Improve Crime
Banned	223	75
Not Banned	107	21

Please indicate whether the data indicates that banning guns is likely to increase or decrease crime rates.

Results show that ideology drove the ability to find the right answer.

Cognitive Bias

A cognitive bias is a systematic pattern of deviation from rationality. Individuals create their own "subjective reality" from their perception of the input. An individual's construction of reality, not the objective input, may dictate their behavior in the world.

Some Common Cognitive Bias: [Click here for the whole picture](#)

- Confirmation Bias (discard data that contradicts our beliefs)
- Overconfidence Bias (Dunning-Kruger)
- Anchoring Bias (overrate first impression)
- Clustering Illusion/Apohenia (find unreasonable patterns)
- ...

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- We jump into conclusions with too little data
- We do not revise well our beliefs when confronted with contradictory data
- We overestimate our skills when we shouldn't
- Cognitive bias are unavoidable, but we can mitigate them.

A (utopic) goal for this course

- ① Understand and accept our limitations
- ② Fight our laziness for thinking
- ③ Fight our cognitive biases
- ④ Do "good, independent and useful" thinking

A (utopic) goal for this course

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- 3 Fight our cognitive biases
- 4 Do "good, independent and useful" thinking

S. Sloman (Cognitive Scientist, Brown University)

Most people don't like to think at all, or like to think as little as possible[...]. Even the rest seem to **devote a lot of their resources to justifying beliefs that they want to hold**, as opposed to forming credible beliefs based only on facts and knowledge.